

DNN 24-Hour Exam Master Plan

Based on analysis of uploaded question papers (2024, EC2, EC3), repeated topics are prioritized.
Goal: solve questions independently in exam.

High Priority Topics

- 1 Perceptron and Learning Rule
- 2 Linear Regression + Gradient Descent
- 3 Logistic Regression
- 4 BCE + Sigmoid Derivation
- 5 Softmax + Cross Entropy
- 6 Forward Propagation
- 7 Backpropagation
- 8 CNN Output Shape + Pooling
- 9 Transfer Learning
- 10 CNN vs ANN
- 11 Activation Functions
- 12 RNN/GRU
- 13 Attention
- 14 Transformers
- 15 Optimizers + Regularization

24-Hour Schedule

Hours 1-3: Perceptron, linear regression, gradient descent. Practice weight updates and MSE.

Hours 3-6: Logistic regression, sigmoid, BCE. Memorize $\text{gradient} = (\hat{y} - y)x$.

Hours 6-9: Softmax, CCE, confusion matrix, precision/recall.

Hours 9-12: Forward propagation through 2-layer DFNN.

Hours 12-15: Backpropagation derivations and chain rule.

Hours 15-17: CNN output sizes, pooling, parameter counts.

Hours 17-18: Transfer learning, CNN vs ANN.

Hours 18-20: Activation functions, dropout, batch norm.

Hours 20-22: RNN, GRU, vanishing gradients.

Hours 22-23: Attention and Transformers.

Hour 24: Full mock exam without notes.

Golden Formulas

Must Memorize:

1. $z = Wx + b$
2. $y_{\text{hat}} = \text{sigmoid}(z)$
3. $dL/dz = y_{\text{hat}} - y$
4. $\text{gradient} = (y_{\text{hat}} - y)x$
5. $w_{\text{new}} = w - \eta * \text{gradient}$
6. Softmax: $\exp(z_i) / \sum(\exp(z_j))$
7. CNN Output = $(N-F+2P)/S + 1$

Theory Questions to Memorize

- Why CNN over ANN? -> Parameter sharing, local features, translation invariance.
- Why transfer learning? -> Less data, faster convergence, better generalization.
- Overfitting fixes -> Dropout, regularization, data augmentation, early stopping.
- Vanishing gradients fixes -> ReLU, residual connections, batch norm, GRU/LSTM.

Exam Checklist

- ☐ Solve 3 perceptron questions
- ☐ Solve 3 logistic regression updates
- ☐ Solve 3 CNN output problems
- ☐ Solve 2 backprop derivations
- ☐ Solve 2 softmax problems
- ☐ Solve 1 RNN/GRU problem
- ☐ Solve 1 attention problem